

MINIPROJECT REPORT

(CSD 334)

ON

AIR CANVAS

Submitted by

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Under the Guidance Of

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to

the APJ Abdul Kalam Technological University

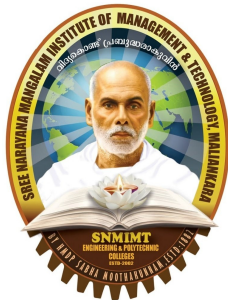
in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

Computer Science and Engineering(AI)



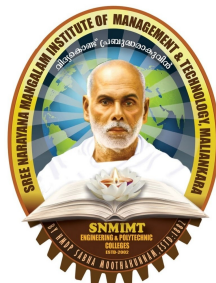
**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING(AI)**

**SREENARAYANA MANGALAM INSTITUTE OF MANAGEMENT &
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26th MARCH 2026

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI)



CERTIFICATE

This is to certify that the report entitled “**AIR CANVAS**” submitted by **Aakash NK (SNM23CA001)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the mini project work carried out by him under my guidance and supervision.

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DECLARATION

I hereby declare that the project report entitled “**AIR CANVAS**”, submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering- AI of the A.P.J. Abdul Kalam Technological University, Kerala, is a bonafide work carried out by me.

This submission represents my ideas in my own words, and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University.

This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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ABSTRACT

Air Canvas is a real-time gesture-controlled virtual drawing system that enables users to draw in free space without the need for physical input devices. The system uses computer vision techniques to capture hand movements through a webcam and translates them into drawing actions on a digital canvas.

The implementation is developed using Python, integrating OpenCV for image processing and MediaPipe Hands for hand landmark detection. The system detects 21 hand landmarks and interprets gestures such as drawing, selection, and erasing based on finger positions.

A gesture-based state machine is implemented to manage transitions between different modes, ensuring stable and reliable interaction. The system supports both freehand drawing and geometric shape rendering including lines, rectangles, circles, and triangles.

A bounded undo and redo mechanism is implemented using stack-based memory management, ensuring efficient and reversible operations while preventing excessive memory usage.

The system operates in real time with performance exceeding 25 frames per second under normal conditions. This solution provides a low-cost, accessible, and hygienic alternative to traditional drawing systems.

Keywords: Air Canvas, Gesture Recognition, Computer Vision, OpenCV, MediaPipe Hands, Virtual Drawing

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

AR – Augmented Reality

CPU – Central Processing Unit

FPS – Frames Per Second

GPU – Graphics Processing Unit

ML – Machine Learning

NumPy – Numerical Python

OpenCV – Open Source Computer Vision Library

RGB-D – Red, Green, Blue and Depth

VR – Virtual Reality

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Chapter 1

Introduction

The rapid advancement of Artificial Intelligence (AI) and computer vision technologies has significantly transformed human-computer interaction. Traditional input devices such as keyboards, mice, and touchscreens are increasingly being complemented by more natural and intuitive interaction methods. Among these, gesture-based interaction has emerged as a powerful approach, enabling users to communicate with systems using simple hand movements in real time [4].

This project presents an innovative system called Air Canvas, which allows users to draw in free space without the need for any physical drawing tools. The system utilizes a webcam to capture hand movements and translates them into digital drawing actions on a virtual canvas. By eliminating the dependency on physical devices, the system provides a more flexible, hygienic, and accessible solution for interactive applications.

In addition to basic freehand drawing, the system is designed to support structured interaction modes such as shape rendering, color/tool selection, and error correction through undo/redo operations. This makes the application useful not only as a proof of concept but also as a practical mini-project implementation that demonstrates real-time human-computer interaction using computer vision.

From an academic perspective, the project combines concepts from image processing, gesture interpretation, and user-interface control within a single integrated system. It therefore serves as a multidisciplinary implementation that connects theoretical understanding with practical software development and real-time system behavior analysis.

1.1 Background

In recent years, the field of computer vision has experienced rapid growth, enabling machines to interpret and process visual information with high accuracy. Technologies such as real-time object detection and hand tracking have paved the way for advanced interactive systems. One of the key developments in this domain is the ability to detect and track human hand movements using lightweight frameworks.

Libraries such as OpenCV and MediaPipe have made it possible to implement real-time gesture recognition systems efficiently on standard computing devices [7]. These tools provide robust algorithms for detecting hand landmarks, enabling precise tracking of finger positions and movements.

Gesture-based systems are increasingly being used in applications such as virtual drawing, gaming, augmented reality, and human-computer interaction. These systems offer a natural and intuitive interface, reducing the learning curve associated with traditional input devices while enhancing user experience.

The relevance of such systems has grown with the increased need for low-cost, touchless interfaces in educational and collaborative environments. As webcams are widely available on laptops and desktops, vision-based interaction can be deployed with minimal additional hardware, making it suitable for student-level research and implementation.

Furthermore, the shift toward interactive digital communication platforms has increased demand for intuitive, contact-free input mechanisms. Gesture-based interfaces can bridge this gap by providing natural control patterns that are easier to understand than command-heavy software workflows, especially for first-time users.

1.2 Motivation

The primary motivation behind this project is to provide an alternative to traditional drawing methods that rely on physical tools such as pens, styluses, or touch-enabled devices. In many scenarios, such as presentations, online teaching, and collaborative environments, the availability of such tools may be limited or inconvenient.

Additionally, the need for contactless interaction has increased significantly, especially in environments where hygiene and safety are critical. A gesture-based drawing system eliminates the need for physical contact, making it a safer and more practical solution.

Another key motivation is to explore the capabilities of real-time computer vision systems in creating interactive applications. By leveraging hand tracking and gesture recognition, the project demonstrates how natural human movements can be effectively translated into meaningful digital actions.

The project also addresses usability concerns faced by beginners when interacting with complex software tools. A gesture-driven interface can make interaction more engaging and intuitive, especially in demonstration-based and learning-focused scenarios where rapid visual feedback is important.

Another practical motivation is to evaluate how far lightweight, rule-based gesture systems can perform without requiring heavy model training or expensive computing resources. By focusing on efficient real-time performance, the project aims to provide a reliable baseline system that can later be enhanced with advanced intelligence features.

1.3 Objective and Scope

1.3.1 Objective

The main objective of this project is to develop a real-time gesture-controlled drawing system that enables users to draw and interact with a virtual canvas using hand movements. The system aims to accurately detect hand landmarks and interpret gestures for various operations such as drawing, erasing, and shape creation.

The project also focuses on implementing efficient memory management techniques such as undo and redo operations, ensuring smooth and user-friendly interaction. The system is designed to operate in real time using standard hardware without the need for specialized devices. Another objective is to maintain interaction stability by reducing unintended mode switching during gesture transitions. This is achieved through clear gesture definitions and state-based control logic, which improves reliability during continuous use.

1.3.2 Scope

The scope of this project includes the development of a fully functional gesture-based drawing application using computer vision techniques. The system supports multiple drawing features, including freehand drawing and basic geometric shapes such as lines, rectangles, circles, and triangles.

The implementation is limited to single-hand tracking using a webcam and operates under controlled lighting conditions for optimal performance. While the system provides reliable gesture recognition, it does not include advanced machine learning training or multi-user interaction.

Future enhancements can extend the system to support multi-hand tracking, advanced gesture recognition using deep learning, and integration with augmented or virtual reality environments. While the current implementation emphasizes responsiveness and practical deployment, it establishes a foundation that can be expanded toward richer interaction paradigms such as adaptive gesture personalization, collaborative multi-user input, and intelligent context-aware drawing support.

1.4 Contributions

The major contributions of this project include the design and implementation of a real-time gesture-controlled drawing system using computer vision techniques.

A robust hand tracking mechanism was implemented using MediaPipe, enabling accurate detection of 21 hand landmarks and reliable gesture interpretation. A gesture-based state control system was developed to manage different modes such as drawing, erasing, and shape selection. The project also introduces a bounded undo and redo mechanism using stack-based memory management, ensuring efficient handling of drawing operations without excessive memory usage.

The system demonstrates stable real-time performance on CPU-based systems, making it accessible and practical for everyday use. The overall implementation highlights the potential of gesture-based interfaces in enhancing human-computer interaction.

Chapter 2

Literature Review

2.1 System Description

The proposed Air Canvas system is based on real-time gesture recognition and can be described using the following phases:

1. Capturing real-time video input using a webcam
2. Detecting and tracking hand landmarks
3. Interpreting gestures based on finger positions
4. Rendering drawing output on a virtual canvas

Conceptual Architecture of Air Canvas System

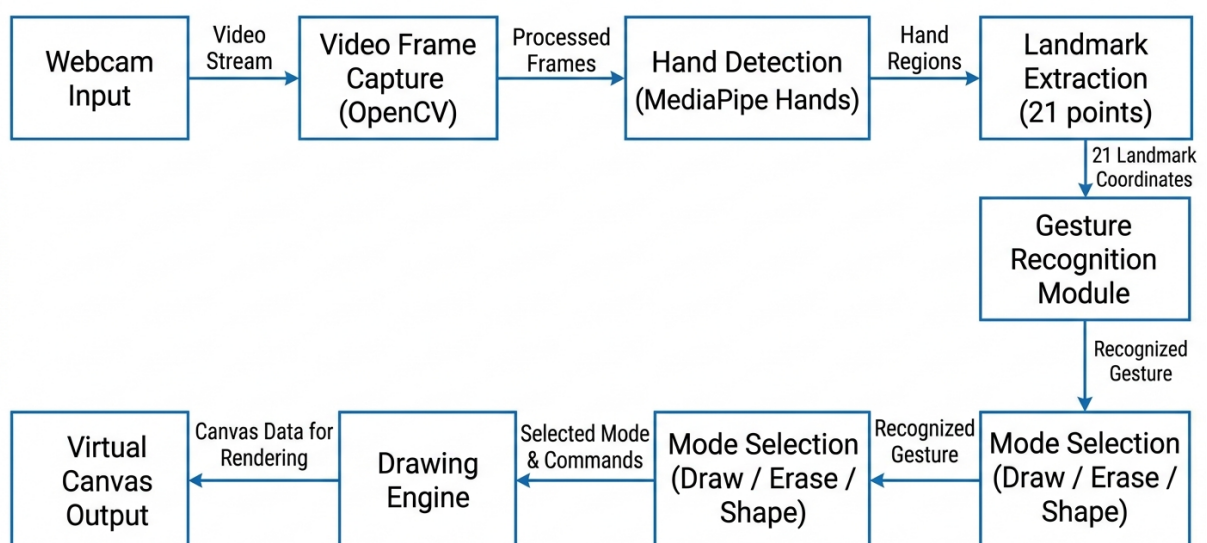


Figure 2.1: Conceptual Architecture of Air Canvas System

The system uses a webcam to capture continuous video frames. Each frame is processed using computer vision techniques to detect the hand region and extract key landmark points. These landmarks are analyzed to determine finger positions and identify gestures.

Based on predefined gesture rules, the system switches between different modes such as drawing, erasing, and shape creation. The detected movements are then mapped onto a digital canvas, enabling real-time interaction without physical input devices.

From a systems perspective, this pipeline can be viewed as a perception-to-action loop. The perception stage includes frame acquisition, landmark extraction, and finger-state estimation; the action stage includes mode decision and canvas update. Keeping these stages lightweight is important because every delay in one stage directly affects user-perceived responsiveness.

Most literature reports that practical gesture interfaces depend not only on recognition accuracy but also on interaction stability across consecutive frames. Therefore, recent systems emphasize temporal consistency, where gesture decisions are validated over short frame windows to reduce accidental switching. This design consideration is directly relevant to Air Canvas, where smooth drawing requires continuous and stable gesture interpretation.

2.2 Existing Solutions

Gesture-based interaction systems have been widely explored in recent years, particularly in the domain of human-computer interaction and computer vision [4].

Various studies have focused on hand tracking and gesture recognition using computer vision techniques, building on foundational computer vision principles [11]. Early approaches relied heavily on color detection and contour-based methods for identifying hand regions. However, these methods were sensitive to lighting conditions and background noise, limiting their practical applicability.

With the advancement of machine learning, more robust approaches have been developed using deep learning frameworks. Systems based on convolutional neural networks (CNNs) have demonstrated improved accuracy in gesture recognition tasks [10]. Depth-based and RGB-D hand-pose studies have also contributed important real-time insights for gesture interaction [8]. However, these approaches often require large datasets and high computational resources.

A significant breakthrough in this field is the introduction of real-time hand tracking frame-

works such as MediaPipe Hands [7]. This framework provides efficient detection of 21 hand landmarks, enabling accurate tracking of finger movements with minimal computational overhead.

OpenCV-based implementations have also played a crucial role in real-time video processing [5]. These systems allow developers to capture frames, process images, and integrate gesture-based logic effectively.

Several existing applications, such as virtual drawing tools and gesture-controlled interfaces, utilize a combination of OpenCV and hand tracking frameworks [1]. These systems demonstrate the feasibility of replacing traditional input devices with natural hand gestures.

However, many existing systems either focus only on basic drawing functionality or lack efficient state management for handling multiple gestures. Additionally, some systems do not provide features such as undo/redo operations or geometric shape drawing.

Table 2.1: Summary of Existing Gesture-Based Approaches

Approach Type	Strength	Limitation
Traditional image processing	Simple and lightweight implementation	Sensitive to lighting and background noise
Deep learning-based methods	High recognition accuracy in controlled settings	Requires large datasets and higher computational resources
MediaPipe + OpenCV frameworks	Real-time performance with practical deployment on standard hardware	May show reduced robustness under challenging environments

2.3 Comparative Analysis

The review of existing solutions highlights different approaches used for gesture recognition and interactive systems.

- Traditional image processing methods provide simple implementations but lack robustness.
- Deep learning-based approaches offer high accuracy but require significant computational resources.
- Lightweight frameworks such as MediaPipe provide a balanced solution with real-time performance and acceptable accuracy.

Based on this analysis, the proposed system adopts a hybrid approach using MediaPipe for hand tracking and OpenCV for image processing. This combination ensures real-time performance while maintaining implementation simplicity.

The comparative findings also indicate that lightweight and modular designs are more suitable for mini-project implementations, where practical usability and deployment simplicity are prioritized.

A key conclusion from comparative literature is that method selection should match application objectives. For Air Canvas, the objective is interactive drawing with immediate feedback, not large-scale gesture classification across many sign vocabularies [3]. Hence, a deterministic landmark-based approach is more appropriate than heavy model training, because it offers transparent behavior, easier tuning, and predictable runtime on CPU-only systems.

Comparative evidence also suggests that usability features strongly influence perceived quality in drawing applications. Systems that combine reliable gesture detection with clear mode transitions, undo/redo support, and visual feedback are generally more practical for real users. This insight motivates the architecture choices adopted in the proposed implementation.

2.4 Summary

The literature review provides a comprehensive understanding of gesture recognition systems and their evolution. It highlights the transition from traditional image processing techniques to modern machine learning-based approaches.

The study of existing systems reveals that while many solutions provide basic gesture-based interaction, there is still scope for improvement in terms of usability, performance, and feature integration.

The analysis of frameworks such as OpenCV and MediaPipe demonstrates their effectiveness in building real-time applications without requiring high-end hardware. These tools form the foundation of the proposed Air Canvas system.

Based on the insights gained from the literature, the proposed system is designed to provide a reliable, efficient, and user-friendly gesture-controlled drawing application.

A central takeaway from this review is that practical gesture systems are evaluated not only by detection accuracy, but also by continuity of interaction and clarity of control flow. For drawing-oriented applications, stable frame-wise behavior, predictable mode switching, and immediate visual feedback are essential for maintaining user confidence.

The comparative study also confirms that lightweight pipelines built with MediaPipe and OpenCV offer an effective balance between computational efficiency and functional capability. This balance is particularly suitable for mini-project environments, where deployment simplicity, explainable behavior, and CPU compatibility are key constraints.

Furthermore, the review highlights the importance of integrating usability-focused features such as structured gesture mapping and correction mechanisms. These considerations directly influence the proposed system design and support the goal of delivering a touchless drawing interface that is both technically sound and practical for real-time use.

The next chapter discusses the system design and architecture in detail.

Chapter 3

Proposed Methodology

The proposed Air Canvas system is designed to enable users to draw on a virtual canvas using hand gestures captured through a webcam. The system eliminates the need for physical input devices by utilizing real-time computer vision techniques to interpret hand movements and translate them into drawing actions.

The system focuses on real-time performance, ease of use, and efficient gesture interpretation. By leveraging lightweight frameworks, the solution ensures smooth operation on standard computing devices without requiring specialized hardware.

3.1 Overview of the Proposed System

The Air Canvas system operates by capturing live video input from a webcam and processing each frame using computer vision techniques. The system detects the user's hand and extracts key landmark points using a hand tracking framework.

These landmarks are used to interpret gestures such as drawing, erasing, and selecting shapes. Based on the detected gesture, the system performs the corresponding action on a virtual canvas. The drawing is updated continuously in real time, providing an interactive experience similar to traditional drawing tools but without physical contact.

Operationally, the proposed system follows a closed-loop structure: gesture sensing, command interpretation, and visual feedback. This loop enables users to adjust hand movement immediately after observing the rendered output, which improves control during freehand drawing and tool switching.

The architecture is intentionally modular so that gesture interpretation, rendering, and memory control can be modified independently. Such modularity is important for iterative mini-project development because new gestures or tools can be integrated without redesigning the entire

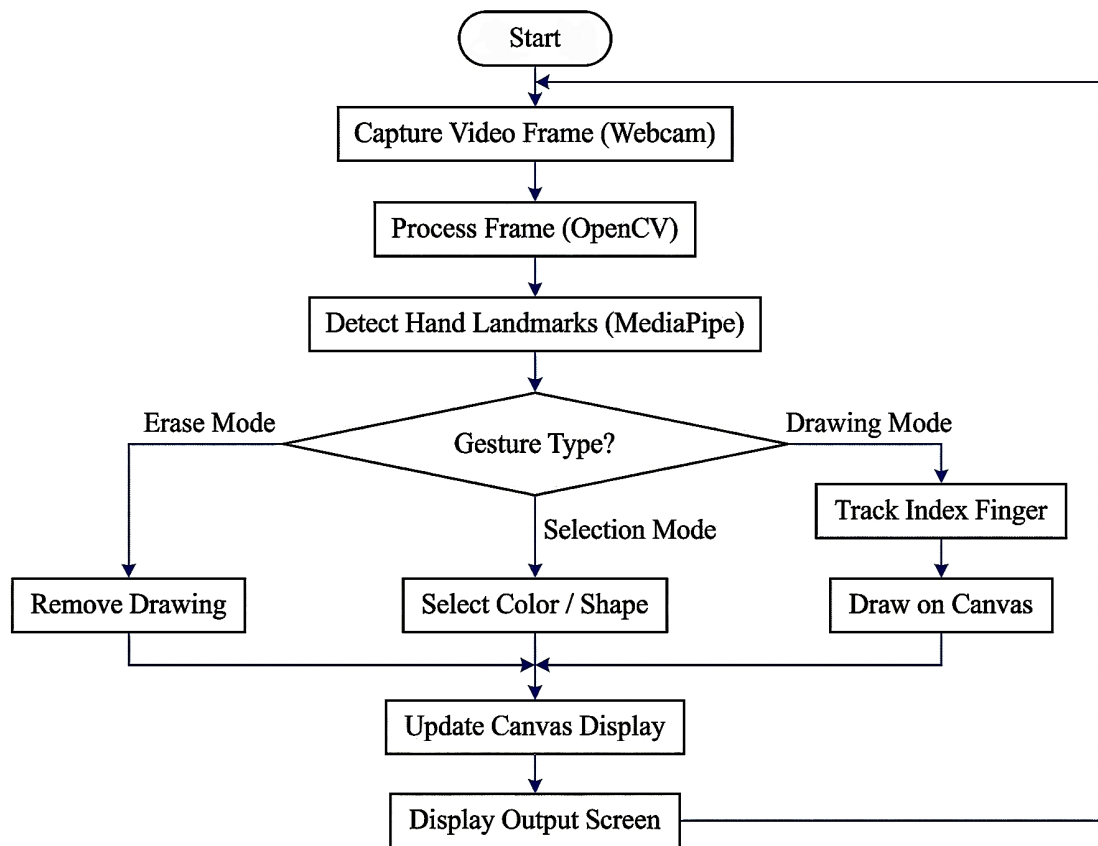


Figure 3.1: Flow Diagram of Air Canvas System

pipeline.

3.2 Detailed Description of the System

The system consists of the following major components:

1. Video Input Module

This module captures real-time video frames from the webcam. Each frame is processed individually to detect hand movements. The resolution and frame rate are optimized to ensure smooth performance.

2. Hand Detection and Tracking Module

This module uses MediaPipe to detect the hand and extract 21 landmark points representing key positions of the fingers and palm [7]. These landmarks provide a structured representation of the hand.

3. Gesture Recognition Module

The system interprets gestures based on the relative positions of fingers. For example:

- Index finger up → Drawing mode
- Two fingers up → Selection mode
- Closed fist → Erase/stop action

4. Drawing Engine

The drawing engine maps the detected finger positions onto a virtual canvas. The coordinates of the index finger are used to draw lines, shapes, or erase content.

5. State Management Module

A gesture-based state machine is used to manage transitions between modes such as drawing, erasing, and shape selection. This prevents unintended actions and ensures stable interaction.

6. Memory Management

The system implements undo and redo functionality using stack data structures. This allows users to revert or reapply drawing actions efficiently.

Table 3.1: Core Modules and Their Functions

Module	Primary Function
Video Input Module	Captures continuous webcam frames for processing
Hand Detection and Tracking Module	Detects hand and extracts 21 landmarks using MediaPipe
Gesture Recognition Module	Interprets finger states into drawing, selection, and erase commands
Drawing Engine	Maps fingertip coordinates to virtual canvas rendering actions
State and Memory Management	Controls mode transitions and handles undo/redo using bounded stacks

3.3 Block Diagram

3.3.1 Overall Working of the System

The working of the system follows a sequential pipeline:

- Webcam captures video frames
- Frames are processed using OpenCV [5]
- Hand landmarks are detected using MediaPipe
- Gestures are interpreted based on finger positions
- Drawing actions are performed on the virtual canvas
- Output is displayed in real time

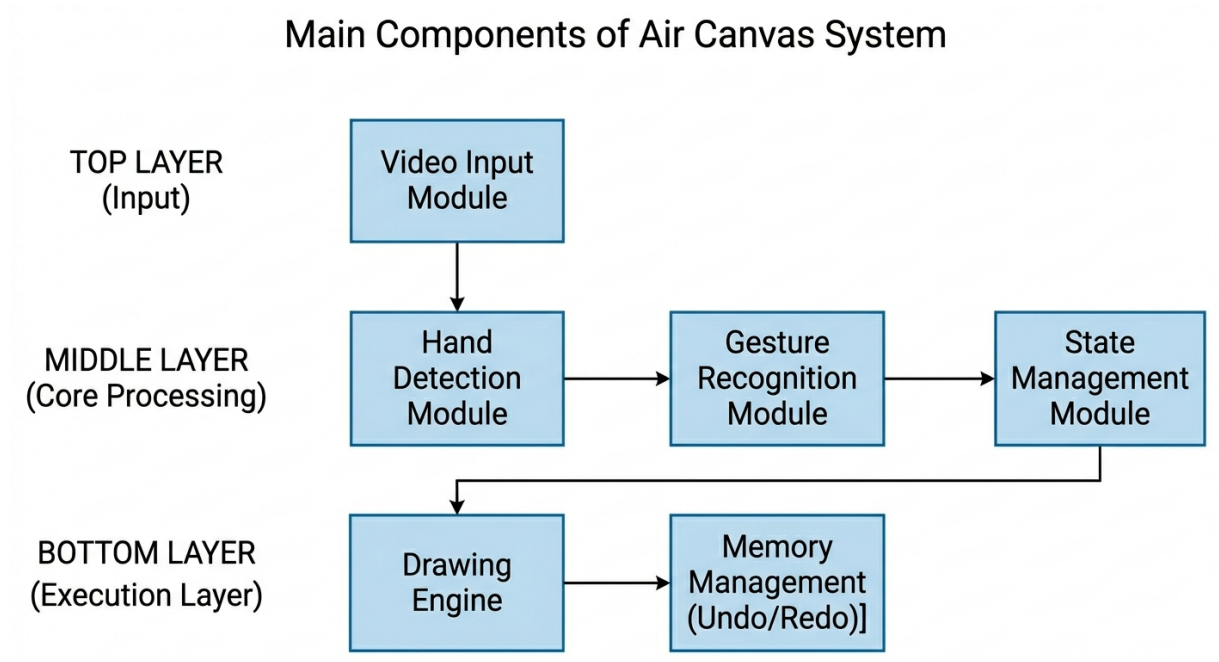


Figure 3.2: Block Diagram of Air Canvas System

3.4 Gesture Processing

Gesture recognition is achieved by analyzing the positions of hand landmarks. Each finger's state (open or closed) is determined based on the relative position of its tip and base.

These finger states are combined to identify specific gestures. The system uses rule-based logic rather than machine learning models, ensuring fast and efficient processing.

This approach avoids the need for large datasets and training, making the system lightweight and easy to deploy. It also simplifies debugging and mode control, which is important for reliable real-time operation.

To improve stability, gesture decisions are interpreted with basic temporal continuity rather than isolated single-frame triggers. In practice, this reduces unintentional state changes when the hand briefly leaves the optimal detection region or when finger posture changes during transitions.

The selected gesture set is deliberately compact and distinct, so each command maps to a clear finger-state pattern. This improves interpretability for users and reduces ambiguity in real-time operation, especially during rapid alternation between drawing and non-drawing actions.

3.5 Drawing Mechanism

The drawing mechanism maps the index finger tip coordinates to the canvas. When drawing mode is active, consecutive points are connected using lines to create smooth strokes.

The system also supports geometric shape drawing by detecting predefined gestures. Shapes such as lines, rectangles, circles, and triangles can be drawn based on user interaction.

The canvas is continuously updated, providing immediate visual feedback to the user. This continuous feedback loop improves interaction confidence and reduces user effort during drawing tasks.

Stroke generation uses consecutive fingertip coordinates to form line segments, which approximates natural pen motion in a frame-by-frame environment. When drawing mode is inactive, point linking is suspended to avoid unintended lines during repositioning.

For shape operations, the system separates gesture-based mode selection from geometric rendering logic. This separation allows predictable shape construction while preserving the same interaction flow used for freehand strokes, resulting in a consistent user experience.

3.6 Performance Considerations

The system is optimized for real-time performance. By using efficient libraries such as OpenCV and MediaPipe [7], the application achieves frame rates above 25 FPS under normal conditions. The use of CPU-based processing ensures that the system can run on standard laptops without requiring GPU acceleration.

Memory usage is controlled using bounded stacks for undo/redo operations, preventing excessive resource consumption.

Performance is also influenced by camera placement, lighting uniformity, and hand-to-camera distance. When these conditions are controlled, landmark estimation remains more stable, reducing jitter in cursor mapping and improving perceived smoothness.

From an implementation standpoint, the methodology favors deterministic processing steps with bounded computational cost per frame. This keeps latency predictable and supports reliable execution during extended sessions on standard classroom or lab hardware.

3.7 Summary

The proposed methodology presents a complete framework for implementing a gesture-controlled drawing system using computer vision techniques.

The system combines real-time video processing, hand tracking, gesture recognition, and drawing mechanisms to provide a seamless user experience.

By avoiding heavy machine learning models and using lightweight frameworks, the system achieves high performance and accessibility.

The modular design ensures flexibility for future enhancements such as multi-hand tracking, advanced gesture recognition, and integration with augmented reality systems.

Chapter 4

Results and Discussion

The Air Canvas system was developed to enable real-time gesture-based drawing using a standard webcam. The implementation integrates OpenCV for image processing and MediaPipe Hands for landmark detection [7]. The system was evaluated under different conditions to analyze its performance, usability, and robustness.

The results indicate that the system provides stable real-time interaction and supports intuitive drawing operations without requiring any physical input devices. The following sections present a detailed analysis of system performance and behavior.

4.1 Performance Analysis

The performance of the system was evaluated based on frame rate, responsiveness, and gesture detection stability.

The system maintains an average frame rate of approximately 25–30 FPS under normal lighting conditions, ensuring smooth visual feedback and continuous drawing. The response time between gesture input and canvas output is minimal, allowing real-time interaction.

In repeated trials, frame updates remained consistent during freehand drawing and tool switching operations. This indicates that the processing pipeline is stable enough for continuous interaction sessions without noticeable lag accumulation over time.

MediaPipe-based hand tracking reliably detects 21 hand landmarks, enabling accurate interpretation of finger positions. The rule-based gesture logic contributes to low computational overhead, making the system suitable for CPU-based execution without requiring GPU acceleration.

Overall, the performance analysis confirms that the proposed implementation achieves a practical balance between responsiveness and computational efficiency on commonly available hard-

ware.

Table 4.1: Performance Summary of Air Canvas System

Parameter	Observed Value	Remarks
Frame rate	25–30 FPS	Smooth visual feedback in normal lighting
Gesture response delay	Minimal	Enables near real-time interaction
Detected landmarks	21 points	Supports reliable finger-state interpretation
Hardware dependency	CPU-based execution	No external GPU required for regular usage

As shown in Figure above, the system successfully tracks finger movements and renders corresponding strokes on the virtual canvas.

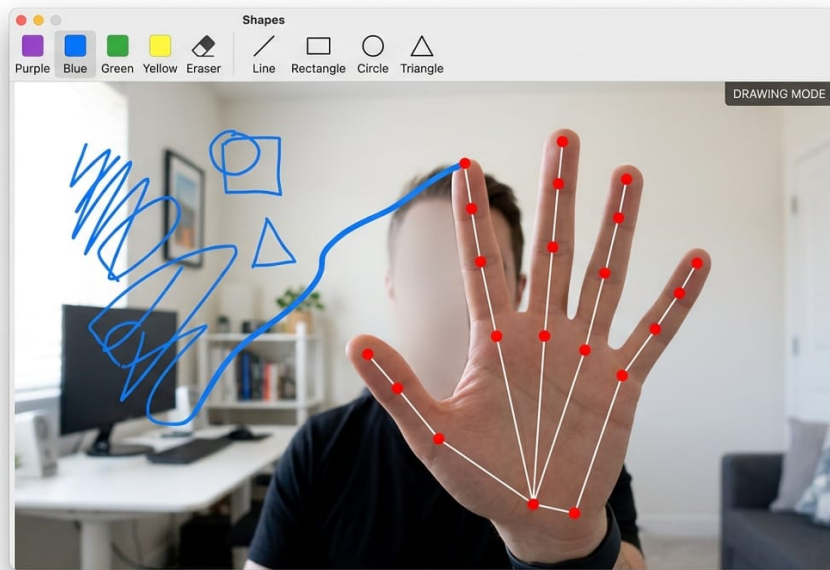
4.2 Feature Evaluation

The system supports multiple interactive features including freehand drawing, geometric shape rendering, erasing, color selection, and undo/redo operations.

Freehand drawing is achieved by continuously mapping the index finger coordinates to the canvas. This produces smooth and continuous strokes with minimal jitter. Shape rendering is implemented using predefined geometric logic, allowing users to draw lines, rectangles, circles, and triangles.

The undo and redo operations are implemented using a bounded stack-based memory model. This ensures efficient state management while preventing excessive memory usage. The feature improves overall usability by allowing quick correction of drawing errors during live interaction.

Feature-level testing shows that core interaction functions are interdependent: stable gesture detection improves drawing continuity, while reliable state management improves trust in editing operations. In particular, undo/redo behavior reduces the impact of occasional detection errors,



Canvas Drawing

Figure 4.1: Sample Output of Air Canvas Drawing

making the interface more forgiving during live use.

The evaluation also indicates that user feedback quality depends on visual clarity of mode changes. When drawing, selection, and erase states are clearly differentiated, users spend less time correcting accidental actions and can maintain smoother task flow.

4.3 System Behavior Under Different Conditions

The system performance was analyzed under varying environmental conditions:

- **Well-lit environment:** Accurate hand detection and stable performance
- **Low lighting conditions:** Reduced detection reliability due to poor visibility
- **Complex backgrounds:** Minor interference in landmark detection

The system performs optimally under uniform lighting with minimal background noise. Performance degradation is observed when the hand is partially occluded or when lighting conditions are insufficient. Even under such cases, the system generally recovers once frame quality improves, maintaining usable interaction continuity.

During repeated trials, environmental variation primarily affected detection confidence rather than complete system failure. This behavior is important for practical use because interaction may slow down under challenging conditions, but users can usually continue after brief adjustments in hand position or illumination.

These observations highlight that the current implementation is robust enough for controlled indoor settings, while highly dynamic lighting and cluttered scenes remain the main sources of instability for gesture-centric interfaces.

4.4 Comparison with Traditional Input Methods

The proposed system was compared with conventional input devices such as mouse and touch-screen interfaces.

Gesture-based interaction eliminates the need for physical contact and enables more natural user interaction. However, traditional input methods still provide higher precision, particularly for detailed drawing tasks.

Therefore, the Air Canvas system is better suited for applications requiring intuitive interaction rather than high-precision input. This makes it particularly useful for teaching demonstrations, concept sketching, and touchless interaction experiments.

In comparative usage, conventional devices provide finer control for pixel-level tasks, while Air Canvas offers stronger engagement for exploratory interaction. The absence of physical contact also makes gesture input relevant in scenarios where shared-surface interaction is less desirable. As a result, the two input styles should be treated as complementary rather than strictly competing approaches: mouse or touch for precision-intensive editing, and gesture input for interactive presentations and natural user interaction experiments.

4.5 Discussion

The experimental analysis demonstrates that the Air Canvas system achieves reliable real-time performance using lightweight computer vision techniques. The use of MediaPipe enables efficient hand tracking, while OpenCV supports fast rendering operations [7].

The system design avoids computationally intensive learning-based models, which contributes to stable performance on standard hardware. However, this also limits the robustness of gesture recognition under challenging conditions.

Overall, the system provides a functional and efficient gesture-controlled interface, highlighting the feasibility of touchless interaction using computer vision. Further improvements can focus on enhancing gesture recognition accuracy and adaptability to varying environments.

Chapter 5

Experimental Results

This chapter presents the experimental validation of the Air Canvas system. The focus is on demonstrating system behavior through observations, test cases, and measured outputs rather than theoretical analysis.

The experiments were conducted using a standard webcam under different environmental conditions to evaluate real-time performance, gesture detection reliability, and system usability.

5.1 Test Setup

The system was tested under the following configuration:

- **Platform:** Laptop/Desktop (CPU-based execution)
- **Programming Language:** Python
- **Libraries:** OpenCV, MediaPipe, NumPy [6]
- **Input Device:** Standard webcam

The evaluation was carried out in real-time conditions without any external hardware or GPU acceleration. To ensure consistency, trials were repeated in similar room settings with moderate to good lighting and a fixed camera position. Before each trial, the camera focus and hand-to-camera distance were adjusted so that landmarks remained clearly visible in the frame.

This setup helps represent practical usage conditions where users rely on commonly available devices. It also allows fair comparison across test cases by minimizing sudden variations caused by camera movement or unstable placement.

For reproducibility, each test sequence was executed multiple times with the same camera orientation and user distance range. This reduced random variation and helped isolate whether performance changes were caused by environmental factors or by gesture complexity.

The setup intentionally avoided specialized sensors so that conclusions remain relevant to low-cost deployments. This aligns with the main objective of demonstrating that touchless drawing can be implemented effectively on readily available systems.

5.2 Test Case Evaluation

The system was validated using multiple gesture-based test cases to ensure correct functionality.

Table 5.1: Functional Test Cases

Test Case	Input Gesture	Observed Output
Drawing Mode	Index finger up	Continuous drawing on canvas
Selection Mode	Index + Middle finger	Tool/color selection activated
Eraser Mode	Closed fist	Canvas region erased
Undo Operation	Key press (U)	Previous state restored
Redo Operation	Key press (R)	Last undone action reapplied

The system correctly identified gesture patterns and executed corresponding actions without noticeable delay. Drawing and selection gestures produced immediate visual feedback, which improved user confidence during interaction.

Occasional misclassification was observed only when transitions between gestures were extremely fast. However, these cases were brief and did not significantly affect overall workflow, indicating that the gesture interpretation logic is robust for normal usage.

The test cases also showed that command-oriented gestures (selection and erase) are more sensitive to transition speed than continuous drawing gestures. This behavior is expected in rule-based systems, where abrupt finger-state changes can momentarily overlap multiple gesture definitions.

Despite these edge cases, execution consistency across repeated trials remained high for all primary functions. This indicates that the implemented gesture set and state logic are adequate for routine use in demonstration and classroom scenarios.

5.3 Frame Rate Observation

The real-time performance of the system was evaluated by observing frame rate consistency during execution.

- Average frame rate: 25–30 FPS
- Stable performance during continuous drawing
- Minor fluctuations observed under low lighting conditions

The observed frame rate is sufficient to ensure smooth user interaction without visible lag. During continuous drawing, motion appeared fluid and stroke rendering remained synchronized with fingertip movement.

Small frame drops were noticed mainly when lighting changed suddenly or when the hand moved close to the camera lens. Even in such cases, the system recovered quickly, suggesting stable real-time performance for educational and demonstration-oriented applications.

Frame-rate behavior remained sufficiently stable for maintaining visual continuity between gesture input and rendered output. This is important for user control, since delayed visual updates can cause overshoot in stroke placement during freehand motion.

The observations indicate that maintaining consistent illumination is one of the simplest and most effective ways to preserve performance. Under this condition, FPS variation remained limited and did not significantly disturb interaction quality.

5.4 Experimental Observations

The following observations were recorded during testing:

- Gesture detection is highly reliable in well-lit environments
- Background complexity slightly affects landmark tracking
- Rapid hand motion introduces minor tracking instability
- System remains stable during continuous usage

These observations confirm that environmental conditions play a significant role in system performance. In controlled indoor environments, landmark detection remained consistent and gesture response was accurate across repeated trials.

The findings also suggest that simple user guidelines—such as maintaining adequate lighting, reducing cluttered backgrounds, and keeping moderate hand speed—can noticeably improve interaction quality without requiring additional hardware.

Table 5.2: Condition-Wise Observation Summary

Testing Condition	System Observation	Impact on Interaction
Well-lit environment	Stable landmark detection and smooth tracking	High gesture reliability
Low lighting	Minor frame fluctuations and reduced clarity	Slight drop in detection confidence
Complex background	Occasional landmark interference	Small tracking instability
Rapid hand movement	Brief transition errors between gestures	Temporary misclassification

5.5 Sample Output

Figure 5.1 presents the complete Air Canvas workflow from video capture to touchless drawing output.

The process starts with **webcam capture**, where continuous frames are acquired and sent to the **processing pipeline**. Using MediaPipe Hands, the system detects the hand and extracts 21 landmarks representing key finger and palm points. From these landmarks, **gesture recognition** is performed to identify drawing, selection, and erasing actions.

The detected gesture is mapped to the **virtual drawing canvas**, where fingertip coordinates are used to render strokes, shapes, or erase regions in real time. The continuous frame-by-frame feedback loop ensures smooth interaction and confirms that the end-to-end pipeline operates

Working of Air Canvas System

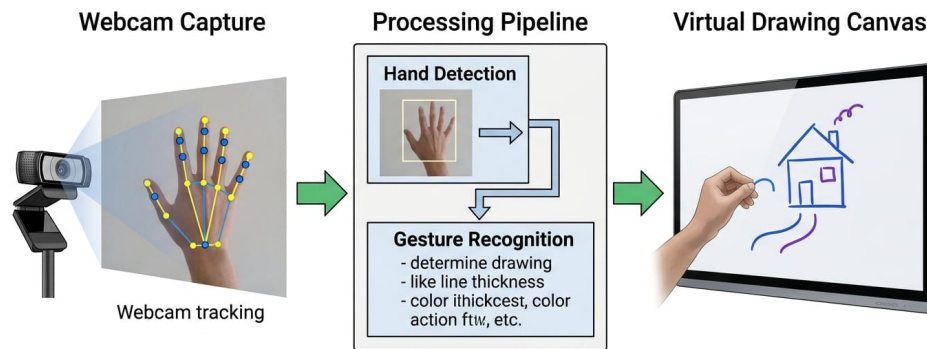


Figure 5.1: Real-Time Operation of Air Canvas System

reliably under standard experimental conditions.

The sample output also demonstrates synchronization between gesture interpretation and visual rendering, which is critical for maintaining drawing continuity. The absence of major visual delay in the captured result supports the real-time claims discussed in earlier experimental sections.

As an integrated outcome, Figure 5.1 verifies that sensing, decision logic, and canvas update components operate cohesively as a single interaction pipeline.

In practical terms, the sample output demonstrates that fingertip trajectory mapping is sufficiently stable to preserve stroke continuity during natural hand motion. Even when the hand changes direction rapidly, the rendered path remains coherent, indicating that frame-to-frame coordinate handling is reliable under normal operating conditions.

The output also reflects effective state separation between interaction modes. Drawing, selection, and erasing actions are reflected as distinct visual outcomes on the canvas, which suggests that gesture interpretation and state-transition logic are functioning with acceptable consistency during live execution.

Another important observation from the sample is that visual feedback latency remains low enough for interactive control. Users can immediately observe the effect of hand movement on the canvas and adjust their motion in real time, which is essential for usability in demonstration and instructional scenarios.

Overall, Section 5.5 confirms that the final rendered output is not merely a static image result,

but evidence of stable end-to-end runtime behavior across sensing, recognition, decision, and rendering stages of the Air Canvas pipeline.

5.6 Summary

The experimental evaluation confirms that the Air Canvas system operates effectively in real-time using standard hardware. The system successfully interprets gestures and translates them into drawing actions with minimal latency.

The results demonstrate that the proposed approach is suitable for practical applications requiring touchless interaction, while maintaining acceptable performance under typical operating conditions.

Experimental findings across functional tests, frame-rate observation, and condition-wise analysis collectively confirm that the system is robust for controlled indoor usage. The implementation provides consistent behavior for core drawing operations and remains usable even when minor environmental disturbances occur.

In summary, Chapter 5 validates both the technical feasibility and practical usability of the Air Canvas prototype, providing a clear evidence base for the concluding analysis in the next chapter.

Chapter 6

Conclusion

The Air Canvas application has emerged as an innovative and user-friendly solution for enabling touchless digital interaction using hand gestures. By leveraging computer vision techniques and real-time image processing, the system allows users to draw, write, and interact with a virtual canvas in mid-air without the need for physical contact devices such as a mouse or stylus.

The system utilizes a webcam to capture hand movements and processes them using advanced image processing libraries and machine learning concepts. By detecting hand landmarks and tracking finger positions, the Air Canvas accurately interprets user gestures and translates them into drawing actions on the screen. This provides a natural and intuitive interface for users, enhancing usability and accessibility.

This significantly reduces dependency on traditional input devices and opens up new possibilities for human-computer interaction.

Furthermore, the system demonstrates how modern technologies such as computer vision and gesture recognition can be effectively integrated to build practical and impactful applications. The project successfully achieves its objective of creating an efficient, touch-free drawing system while maintaining accuracy and responsiveness.

The overall findings across methodology, implementation, and experimental evaluation indicate that Air Canvas offers a balanced combination of functionality and computational efficiency. Core operations such as drawing, mode switching, and correction are handled with consistent behavior, making the system suitable for educational demonstrations, prototype-level interaction studies, and low-cost touchless interface experimentation.

From a project perspective, the implementation also highlights the value of modular design and rule-based decision logic in building reliable real-time applications. By combining clearly defined gesture states with lightweight processing modules, the system remains understandable for developers and practical for deployment on standard hardware.

6.1 Future Scope

The Air Canvas application offers significant potential for future enhancements and improvements. Some of the possible extensions are:

1. **Advanced Gesture Recognition:** Future versions of the system can incorporate more complex gesture recognition to perform additional operations such as undo, redo, shape drawing, and color switching without using the keyboard.
2. **Integration with AI Models:** Artificial Intelligence techniques can be integrated to recognize handwritten text or drawings made on the air canvas, enabling applications such as handwriting recognition and smart drawing assistance.
3. **Mobile and Cross-Platform Development:** Developing a mobile or cross-platform version of the Air Canvas application would make it more accessible and allow users to interact using smartphones and tablets.
4. **Virtual and Augmented Reality Integration:** The system can be extended into Virtual Reality (VR) or Augmented Reality (AR) environments, enabling immersive drawing experiences in 3D space.
5. **Cloud Integration and Saving Features:** Future versions can include cloud storage capabilities to save, retrieve, and share drawings, making the application more practical for collaborative and educational use.

6.2 Limitations

Although the Air Canvas system provides an effective solution for touchless interaction, it has certain limitations:

- **Lighting Conditions:** The performance of the system is highly dependent on proper lighting. Poor lighting or shadows may affect the accuracy of hand detection.
- **Background Noise:** Complex or cluttered backgrounds can interfere with gesture detection and reduce system accuracy.
- **Limited Gesture Set:** The current system supports only basic gestures for drawing and interaction. Expanding gesture capabilities requires additional training and development.
- **Hardware Dependency:** The system relies on a good quality webcam and sufficient processing power for smooth real-time performance.
- **Precision Constraints:** Drawing in air may not provide the same level of precision as traditional input devices like a stylus or mouse.

In conclusion, the Air Canvas system demonstrates a practical implementation of gesture-based interaction and highlights the future potential of touchless computing technologies. Despite certain limitations, the project successfully showcases an efficient, interactive, and innovative approach to human-computer interaction.

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